TOPOLOGICAL INSULATORS — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 16, 2026

Diagram-based questions on Topological Insulators

Question 1

10.0 points

Analyze the band structure in the diagram below of a topological insulator, which features band inversion between the bulk valence and conduction bands. Explain how this inversion leads to a non-trivial topology.

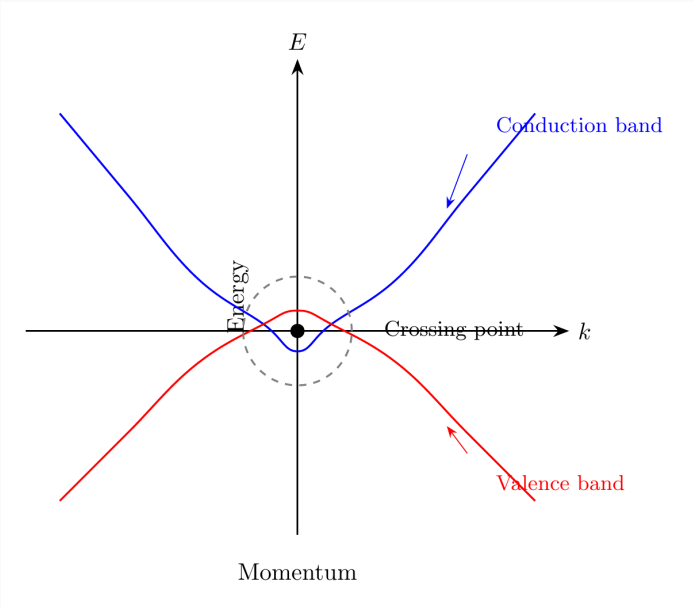

Fig. 1. A band structure diagram of a topological insulator, showing band inversion. The diagram should include the bulk valence band and conduction band, with the valence band curving upwards and the conduction band curving downwards, crossing each other to indicate inversion. The x-axis represents momentum (k), and the y-axis represents energy (E). Highlight the crossing point where the bands invert, indicating the non-trivial topology. Label the valence band, conduction band, and the inversion point clearly. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • z2 topological invariant • band inversion • logical operator • logical qubit encoding path The diagram will FAIL review if any of these appear.

Question 2 10.0 points

Interpret the bulk-boundary correspondence as depicted in the surface state diagram of a topological insulator in the diagram below. Describe how the surface states are protected by topology.

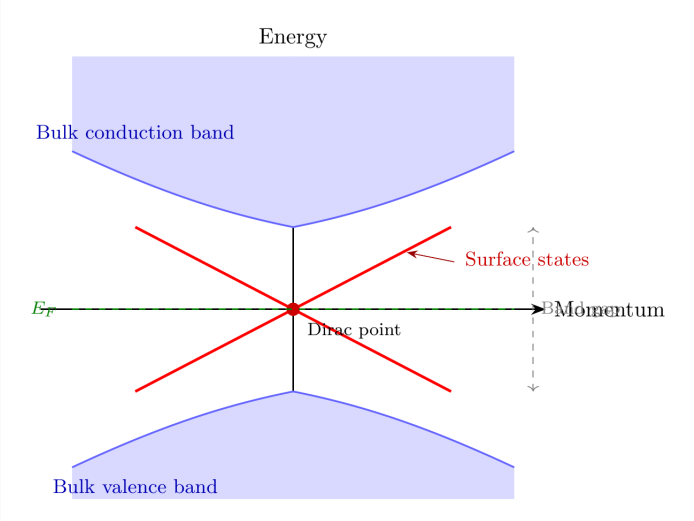

Fig. 2. Surface state diagram of a topological insulator showing bulk bands and surface states. The diagram should include the bulk conduction band and valence band, with a band gap between them. Surface states should be depicted as lines crossing through the band gap, illustrating the topological protection. Label the bulk bands and surface states clearly. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • These • z2 topological invariant • band inversion • logical operator • logical qubit encoding path The diagram will FAIL review if any of these appear.

Question 3 10.0 points

Examine the edge state diagram shown below associated with the quantum spin Hall effect. Explain how these states allow for dissipationless edge transport.

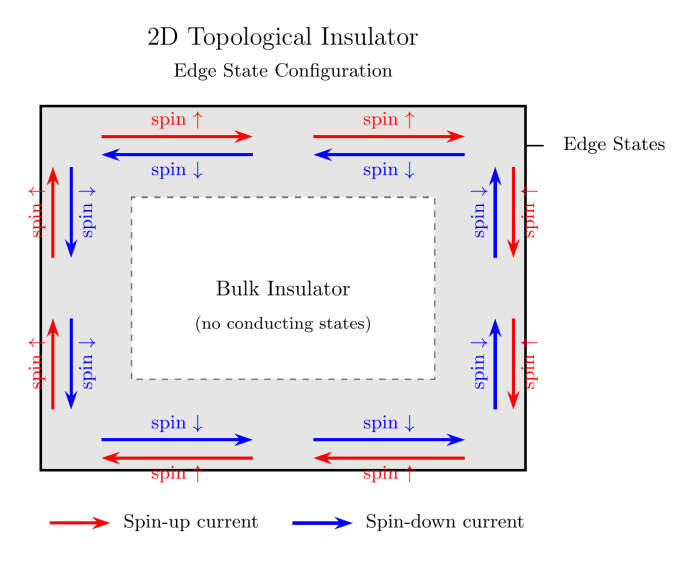

Fig. 3. A quantum spin Hall effect edge state diagram. The diagram should depict a two-dimensional topological insulator with edge states. Show the bulk insulator region in the center with no conducting states, and the edge states on the boundaries. Indicate the direction of electron spin (up and down) along the edges, illustrating how they propagate in opposite directions. Use arrows to show the direction of electron flow along the edges, highlighting the dissipationless nature of transport. Label the bulk insulator and edge states clearly. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • Due • Hall • z2 topological invariant • logical operator • logical qubit encoding path The diagram will FAIL review if any of these appear.

Question 4 10.0 points

Analyze the k-space structure of a 2D system with a Z2 topological invariant as depicted in the diagram below. Explain the significance of the Z2 invariant.

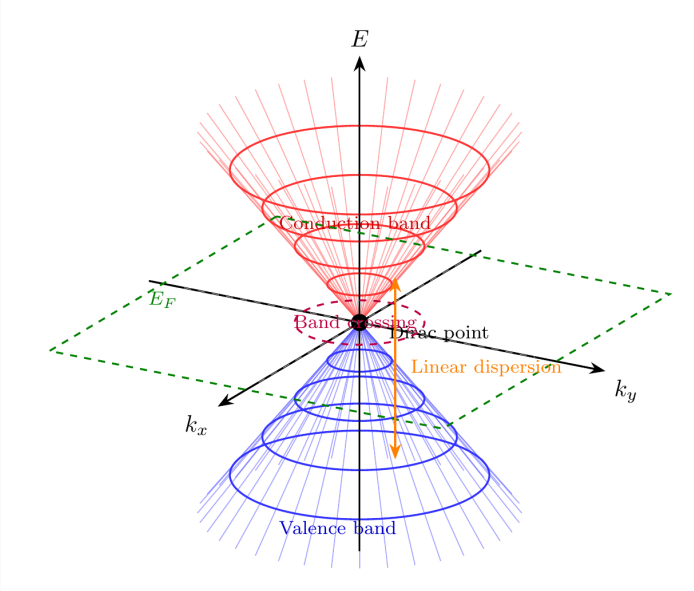

Fig. 4. CRITICAL: The previous diagram showed the wrong type. The diagram shows a top-down view of a 2D Brillouin zone, but the question and description require a band structure diagram (Energy vs. k) that depicts a Dirac cone formed by the crossing of conduction and valence bands.. Specifically: The diagram is of the wrong. Regenerate from scratch showing the CORRECT diagram type. A diagram of the k-space structure for a 2D system with a Z2 topological invariant. The diagram should show a typical band structure with a depiction of the Dirac cone at the boundary of the Brillouin zone, indicating the presence of edge states. The Z2 invariant is represented by a band inversion, where the conduction and valence bands touch at a point, forming a Dirac cone. Label the axes as kx and ky, and highlight the Dirac point where the bands touch. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • z2 topological invariant • logical operator • logical qubit encoding path The diagram will FAIL review if any of these appear.

Question 5 10.0 points

Interpret the Dirac cone dispersion relation in the diagram below on the surface of a 3D topological insulator. Discuss how this structure is related to massless Dirac fermions.

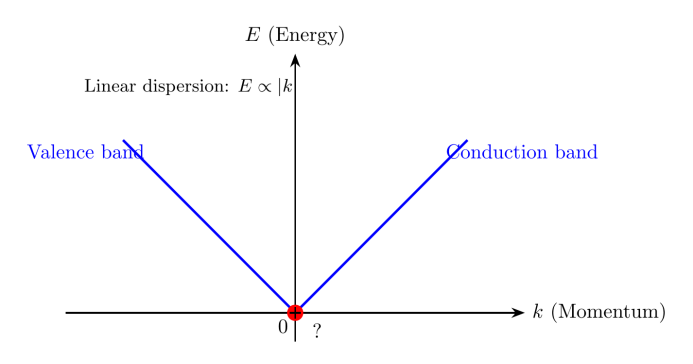

Fig. 5. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: Dirac, z2 topological invariant, logical operator, logical qubit encoding path; The diagram contains the label "Dirac point", which includes the term "Dirac", a forbidden label for this question, thus revealing part of the answer.. WRONG: 3D double-cone structure with E, kx, and ky axes. CORRECT: 2D dispersion relation plot with a V-shape. VISUAL FEATURES: The diagram must be a 2D plot. The vertical axis should be labeled 'E' (Energy). The horizontal axis should be labeled 'k' (Momentum). The plot should show two straight lines forming a 'V' shape, touching at the origin (k=0, E=0), which should be labeled as the 'Dirac point'. DO NOT draw a 3D cone structure. IMPORTANT correction needed: The diagram contains the label "Dirac point", which includes the term "Dirac", a forbidden label for this question, thus revealing part of the answer.. Specifically: The label "Dirac point" is present, but "Dirac" is on the forbidden label list. CORRECTED_DESCRIPTION. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.